National Research University Higher School of Economics
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Programme: Data Science & Business Analytics

BACHELOR'S THESIS

Financial Markets and IT Project

Bond Value Analysis and Market Inefficiency Detection Tool for the Russian Debt Market

Prepared by the student of Group БПАД242, 2nd Year of Study

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Moscow, 2026

Abstract

Bond Value Analysis and Market Inefficiency Detection Tool

My goal is to develop an automated system for analyzing bonds on our Russian Debt Market. My tool collects bond data from financial websites*, calculates theoretical values using discounted cash flow models with various adjustments (reinvestment, taxes, payment frequency), and compares these values with real-time market prices through Tinkoff API integration. Bonds are categorized into comparison groups based on credit rating, coupon type, and other characteristics (forming baskets). The system highlights bonds with price deviations exceeding $5 - 10\%$ from their group averages, indicating potential undervaluation or overvaluation (at that point the user should look into these concrete bonds himself). The Python implementation provides an easy to work with interface with color-coded tables, sorting capabilities, and risk assessment features. The developed tool helps investors make more informed decisions in the Russian debt market by systematically identifying mispriced bond issues and potential investment opportunities, increasing the efficiency of capital use.

* - (https://smart-lab.ru/q/bonds/)

Contents

1 Introduction

1.1 Background

This project develops an automated tool for analyzing Russian bonds and identifying pricing inefficiencies. The system uses mathematical models to calculate bond values and compares them with real-time market prices. This gives investors an analyzed overview of the bond market and helps find bonds that may be incorrectly priced and make better investment decisions.

1.2 Problem Statement

Most individual investors in the Russian bond market do not have access to advanced analysis tools. They cannot easily calculate correct bond values or compare bonds effectively. Even if they do have the knowledge on valuing bonds, it takes a lot of time to go through a large chunk of bond issues. My tool addresses these problems, making it more convenient to analyze the market for uninstitutional investors.

1.3 Relevance

The relevance of this project is explained by current research on algorithmic trading and automation in financial markets. According to recent studies, the use of robotics and algorithmic trading in financial markets is growing in a very fast tempo. Research shows that:

- **Robotics adoption in financial sectors:** Recent studies indicate high adoption rates of robotic systems across financial sectors, with 85% in retail banking, 80% in investment banking, 75% in hedge funds, 70% in asset management, and 65% in insurance. This demonstrates the rapid automation of financial processes.
- **Algorithmic trading dominance:** Research confirms that algorithmic trading has already transformed financial markets, with high-frequency trading (HFT) algorithms executing the majority of trades in developed markets (Russian stock market is not yet considered a developed market, it is now classified at the "emerging" state). These systems can analyze massive amounts of market data and execute trades faster and more accurately than humans.
- **Machine learning integration:** Studies show that machine learning algorithms are being integrated into trading systems (with an increasing tempo), improving trend analysis and prediction accuracy. These systems can identify complex patterns that traditional analysis might easily miss.

- **Future projections:** Based on the trends we can see now, researchers predict that within the next 5-10 years, almost all trading in developed markets will be conducted by automated systems and algorithms. The financial industry is moving toward complete automation, where human decision-making will be supplemented or replaced by algorithmic systems.

This research context makes my project pretty relevant. The idea is to be among the first to apply automated analysis specifically to the Russian bond market. Historical evidence shows that early adopters of technological innovations in financial markets often gain significant advantages. Those who first properly implement information technology for market analysis can earn large profits before these methods become will be adopted by most of the market.

Russian market presents a unique opportunity for several reasons:

- It is less efficiently analyzed than Western markets
- Pricing inefficiencies are more common
- Automated analysis tools are less widespread
- The market is growing and developing rapidly (in particular, the bond market in Russian has become more popular among individual investors in recent years due to high key rate)

By developing this automated bond analysis tool now, investors (and me, personally) can gain an advantage before automated analysis becomes standard practice in the Russian market. This project connects advanced algorithmic robotic techniques used in Western markets and the current state of analysis in the Russian bond market.

1.4 Thesis Tasks

1. Study existing bond valuation methods and algorithmic trading research
2. Develop (apply) mathematical models for bond value calculation
3. Create automated data parsing system from financial websites
4. Build bond comparison system with grouping by characteristics (forming baskets)
5. Design user-friendly interface with visual indicators
6. Test the system with real Russian bond market data (using Tinkoff API)

1.5 Thesis Objectives

The main goal is to create a working automated tool that:

- Identifies bond valuation differences systematically
- Helps investors find investment opportunities

- Works with real-time market data
- Is accessible to individual investors
- Provides clear, useful information in a form of a table
- Is able to analyze a large chunk of bond issues in under 20 seconds

2 Literature Review

Three main areas of research provide the base for my project:

Algorithmic Trading Research: Studies show how algorithmic trading has transformed financial markets. Their research demonstrates increased efficiency, speed, and accuracy in trading execution. However, they also note challenges including increased market volatility and the need for regulatory frameworks.

Bond Valuation Studies: Russian research by Чистикова И.В. ("Методические подходы к оценке рыночной стоимости облигаций") provides basic bond valuation methods. This work explains different approaches including income-based valuation, comparative analysis, and cost methods. Explains the importance of cash flow analysis and adjustments for different payment schedules.

Market Efficiency Research: Studies by Сафонов Д.Ю. ("Особенности применения доходного подхода к оценке стоимости облигаций в условиях рыночных изменений") examine how bond prices react to market changes. This research shows the inverse relationship between bond prices and interest rates, and how sensitivity varies with maturity and coupon characteristics.

Professional Market Methodology: The National Settlement Depository (НКО АО НРД) methodology for determining the value of bonds without international ratings [11] provides a comprehensive framework with three-level hierarchy of valuation methods. This methodology, approved by the NSD Expert Council, establishes procedures for fair value assessment based on market data availability and uses advanced statistical techniques including Nelson-Siegel-Svensson yield curve modeling.

Comparative Model Analysis: Research by Истомин Н.А. [12] provides a systematic review of bond valuation models, comparing structural models, reduced-form models, credit rating-based approaches, neural network models, and spread-to-G-curve methods. This analysis demonstrates the advantages of using issuer's outstanding bond issues for fair value estimation in Russian market conditions.

These studies confirm that automated bond analysis requires both mathematical accuracy and deep understanding of bond valuation principles. The integration of algorithmic approaches with fundamental bond analysis creates new opportunities for investors.

3 Data and Methodology

3.1 Data Collection

The system collects (parses) bond data from two primary sources:

- **SmartLab website:** For comprehensive bond characteristics and market information
- **Tinkoff API:** For real-time market prices and trading data

From SmartLab, the system collects these specific indicators for each bond:

- Лет до погашения
- Доходность
- Годовой купонный доход
- Купонный доход последний
- Рейтинг
- Объем, млн руб
- Купон, руб
- Частота, раз в год
- НКД, руб
- Дюрация, лет
- Цена
- Дата купона
- Размещение
- Погашение
- Оферта

3.2 Key Indicators

The system analyzes bonds using these important factors:

- **Credit rating:** Average from three rating agencies: АКРА (Аналитическое Кредитное Рейтинговое Агентство), Эксперт РА (Рейтинговое Агентство «Эксперт РА») and НКР (Национальные Кредитные Рейтинги).

- **Maturity time:** Grouped into time periods (less than 6 months, 6-12 months, etc.)
- **Coupon type:** Fixed or floating rate
- **Payment frequency:** How often coupon payments are made
- **Special features:** Government ownership, buyback options, gradual repayment, amortization

3.3 Formulas for Bond Valuation

The system uses these mathematical formulas for bond valuation based on the income approach (доходный подход). According to Russian research [6, 7], these formulas account for different bond characteristics and market conditions.

1. Basic formula for annual coupon payments (without reinvestment):

$$S = \sum_{t=1}^{T} \frac{C}{(1+r)^t} + \frac{N}{(1+r)^T} \tag{1}$$

Where:

- S = bond theoretical value (руб.) Constraint: $S > 0$
- C = annual coupon payment (руб.) Source: From SmartLab "Купон, руб" × "Частота, раз в год"
- N = nominal value (руб.) Constraint: $N > 0$, typically 1000 руб. for Russian bonds
- r = discount rate (annual, decimal) Source: Based on Central Bank key rate + risk premium; Constraint: $r > 0$
- T = years to maturity Source: From SmartLab "Лет до погашения"; Constraint: $T \geq 0$

2. Formula with different payment frequency:

If coupon payments occur m times per year (semi-annually, quarterly):

$$S = \sum_{t=1}^{T \cdot m} \frac{C/m}{(1+r/m)^t} + \frac{N}{(1+r/m)^{T \cdot m}} \tag{2}$$

Where:

- m = payment frequency (times per year) Source: From SmartLab "Частота, раз в год"; Constraint: $m \in \{1, 2, 4, 12\}$ (annual, semi-annual, quarterly, monthly)
- C/m = coupon payment per period (руб.)
- $T \cdot m$ = total number of payment periods

3. Formula with coupon reinvestment consideration:

Accounting for reinvestment of received coupon payments [6]:

$$S = \sum_{t=1}^{T} \frac{C \cdot (1 + e \cdot k/K)}{(1+r)^t} + \frac{N}{(1+r)^T} \tag{3}$$

Where:

- e = reinvestment rate (annual, decimal) Source: Bank deposit rate or risk-free rate; Constraint: $0 \leq e \leq 0.20$ (0-20%)
- k = days from coupon receipt to maturity Source: Calculated from coupon dates; Constraint: $0 \leq k \leq 365$
- K = days in reinvestment period Constraint: Typically $K = 365$ for annual calculation

4. Formula with tax adjustment:

Considering personal income tax (НДФЛ) for individual investors:

$$S = \sum_{t=1}^{T} \frac{C \cdot (1 - \tau)}{(1+r)^t} + \frac{N}{(1+r)^T} \tag{4}$$

Where:

- τ = tax rate (decimal) Constraint: $\tau = 0.13$ for Russian residents (13% НДФЛ)
- Note: Only coupon payments are taxed; principal repayment at maturity is tax-free

5. Formula for zero-coupon (бескупонные) bonds:

For bonds without periodic coupon payments [7]:

$$S = \frac{N}{(1+r)^T} \tag{5}$$

Where:

- All variables as defined in equation (1)
- Special case: $C = 0$ (no coupon payments)

6. Formula for perpetual (бессрочные) bonds:

For bonds without maturity date (perpetuities):

$$S = \frac{C}{r} \tag{6}$$

Where:

- C = constant annual coupon payment
- r = required rate of return (capitalization rate)
- Constraint: Only applies when $T \to \infty$

Discount rate (r) determination with credit rating adjustment:

Based on research by [7], the discount rate includes credit risk premium:

$$r = r_f + \pi + \rho(R) \tag{7}$$

- r_f = risk-free rate (ЦБ ключевая ставка) Source: Central Bank of Russia
- π = inflation premium Constraint: Based on expected inflation
- $\rho(R)$ = risk premium depending on credit rating R Constraint: AAA: $0.5-1\%$, AA: $1-2\%$, A: $2-3\%$, BBB: $3-5\%$, BB: $5-7\%$, B: $7-10\%$, CCC and below: 10%+
- Note: For two bonds with identical yield but different credit ratings, the better-rated bond should have a lower discount rate, resulting in a higher theoretical value S

3.4 Basket Formation

Bonds are grouped into "baskets" for comparison:

- By credit rating: AAA, AA, A, BBB, etc.
- By maturity: $<$ 6 months, 6-12 months, 1-2 years, 2-3 years, $>$ 3 years
- By coupon type: Fixed rate, Floating rate
- By currency: Ruble bonds (separate baskets for yuan and dollar bonds)
- By sector*: If sector information is available; * - will not be added in the first versions

3.5 Risk Assessment Methods

- Credit ratings: From three agencies, averaged
- Duration: Measures price sensitivity to rate changes
- Liquidity*: Trading volume as indicator; * - will not be added in the first versions
- Special risks: like option features

4 System Development

4.1 Data Preparation

1. Data collection: Automated collection from SmartLab website
2. Data cleaning: Removing incomplete records, fixing format issues
3. Data validation: Checking for reasonable values
4. Data grouping: Organizing bonds into comparison baskets

The system collects approximately 15 data points for each bond, as shown in the example table with 20 bonds (https://ibb.co/B2q3ZHpK).

4.2 Calculation Engine

1. Basic value calculation: Using the standard discount formula
2. Advanced calculations: With reinvestment and tax adjustments
3. Sensitivity analysis: Testing how values change with rate changes
4. Yield calculations: Different yield measures for comparison

The engine can handle different bond types:

- Regular coupon bonds
- Zero-coupon bonds (no periodic payments)
- Bonds with special features (they will be treated as usual bonds considering the case when no options will be used, but additional risks and options will be identified and highlighted)

4.3 Market Comparison Module

Module compares calculated values with market prices and analyzes yield deviations:

1. Get market prices: From Tinkoff API in real-time
2. Calculate basket averages: Mean values for each bond group including:
   - Average market price in basket
   - Average yield to maturity (YTM) in basket
   - Average duration in basket
   - Average credit rating (converted to numerical score)
3. Compare individual bonds: Each bond vs its basket average using two approaches:

   a) Price-based comparison (short-term opportunities):

$$\Delta_{\text{price}} = \frac{|\text{Calculated Value} - \text{Market Price}|}{\text{Market Price}} \times 100\% \tag{8}$$

   - Focus: Immediate trading opportunities
   - Threshold: $\Delta_{\text{price}} > 5 - 10\%$
   - Suitable for: Short-term trading, arbitrage strategies

   b) Yield-based comparison (long-term investment opportunities):

$$\Delta_{\text{yield}} = \text{YTM}_{\text{bond}} - \text{YTM}_{\text{basket average}} \tag{9}$$

- Focus: Long-term investment value
- Interpretation: Positive Δ_{yield} indicates higher yield than comparable bonds
- Threshold: $|\Delta_{\text{yield}}| > 0.5 - 1.0\%$ (50-100 basis points)
- Suitable for: Buy-and-hold investors, portfolio allocation

4. Combined analysis: Identify bonds with both price and yield anomalies:
   - Undervalued with high yield: $\Delta_{\text{price}} > 5\%$ AND $\Delta_{\text{yield}} > 0.5\%$
   - Overvalued with low yield: $\Delta_{\text{price}} > 5\%$ AND $\Delta_{\text{yield}} < -0.5\%$
   - Fair priced with yield opportunity: $\Delta_{\text{price}} < 3\%$ AND $\Delta_{\text{yield}} > 1\%$
5. Yield curve analysis within basket:
   - Construct yield curve for each basket based on bonds' duration (maturity)
   - Calculate theoretical yield for each bond's duration using curve interpolation
   - Compare actual bond yield with curve-implied yield:

$$\Delta_{\text{curve}} = \text{YTM}_{\text{actual}} - \text{YTM}_{\text{curve}} \tag{10}$$

   - Interpretation: Positive Δ_{curve} indicates bond is above the yield curve (undervalued relative to duration-matched peers), negative indicates below the curve (overvalued)
   - Strength assessment: Calculate z-score of Δ_{curve} relative to basket distribution
6. Risk-adjusted yield comparison with credit rating adjustment:

$$\text{Risk-adjusted Yield} = \frac{\text{YTM}_{\text{bond}} - r_f}{\rho(R)} \tag{11}$$

   - r_f = risk-free rate (ЦБ ключевая ставка)
   - $\rho(R)$ = risk premium for credit rating R
   - Higher values indicate better compensation for credit risk
   - Compare with basket average risk-adjusted yield
   - Important: When two bonds have identical yield but different credit ratings, the formula accounts for their different risk levels, showing which offers better risk-adjusted return

4.4 Visualization Interface

The system presents results in color-coded tables with dual indicators:

- **Color coding for price deviations:**
  - Green: Undervalued by price ($\Delta_{\text{price}} > 5\%$) - good buying opportunity
  - Red: Overvalued by price ($\Delta_{\text{price}} > 5\%$) - possibly overpriced
  - Yellow: Special features need attention
- **Symbol coding for yield opportunities:**
  - $\uparrow\uparrow$: High yield opportunity ($\Delta_{\text{yield}} > 1\%$)
  - $\uparrow$: Moderate yield opportunity ($0.5\% < \Delta_{\text{yield}} \leq 1\%$)
  - $\rightarrow$: Normal yield ($|\Delta_{\text{yield}}| \leq 0.5\%$)
  - $\downarrow$: Low yield ($-1\% \leq \Delta_{\text{yield}} < -0.5\%$)
  - $\downarrow\downarrow$: Very low yield ($\Delta_{\text{yield}} < -1\%$)
- **Sorting options:** Users can sort by:
  - Price deviation (Δ_{price}) - for short-term traders
  - Yield deviation (Δ_{yield}) - for long-term investors
  - Risk-adjusted yield - for risk-aware investors
  - Duration - for interest rate sensitivity analysis
  - Yield curve deviation (Δ_{curve}) - for duration-adjusted opportunities
- **Filtering:** Users can filter by:
  - Investment horizon: Short-term (price focus) vs Long-term (yield focus)
  - Risk tolerance: Credit rating categories
  - Maturity preferences
  - Minimum yield requirements
- **Warning symbols:** Special indicators for bonds with options or other features (e.g. government ownership, call/put options)

4.5 Anomaly Detection

The system finds bonds with unusual pricing using multi-dimensional analysis:

1. **Dual-threshold detection:**
   - Price anomaly: Bonds with $|\Delta_{\text{price}}| > 5 - 10\%$ from basket average
   - Yield anomaly: Bonds with $|\Delta_{\text{yield}}| > 0.5 - 1.0\%$ from basket average
   - Yield curve anomaly: Bonds with $|\Delta_{\text{curve}}| > 2$ standard deviations from basket yield curve
   - Combined anomaly: Bonds meeting multiple criteria (most significant opportunities)

2. **Time horizon classification:**

   Short-term opportunities (trading focus):

   - Primary indicator: Large price deviation
   - Secondary indicator: Liquidity and trading volume
   - Typical holding period: Days to weeks
   - Risk: Market timing, execution risk

   Long-term opportunities (investment focus):

   - Primary indicator: Attractive yield vs comparable bonds
   - Secondary indicator: Credit quality, duration
   - Typical holding period: Months to years
   - Risk: Interest rate changes, credit deterioration
   - Enhanced analysis: Consider position relative to yield curve for given duration

3. **Automatic highlighting:** Bonds are marked with different codes:

   - **P**+: Positive price deviation opportunity
   - **Y**+: Positive yield deviation opportunity
   - **C**+: Above yield curve (duration-adjusted undervaluation)
   - **P** + **Y**+: Both price and yield opportunities (most valuable)
   - **R**: Requires additional research (conflicting signals)

4. **Disclaimers and warnings:**

   - System reminds users that anomalies may have legitimate reasons
   - Users should investigate: Credit events, upcoming news, liquidity issues
   - Historical context: Compare with bond's own historical spreads
   - Market context: Consider overall market conditions and trends

5. **Risk warnings:** Additional checks for:

   - High-risk bonds (BBB and lower ratings)
   - Low liquidity bonds (trading volume $<$ threshold)
   - Bonds with complex features (options, convertibility)
   - Bonds approaching maturity or call dates

6. **Portfolio context analysis:**

   - Consider bond's correlation with existing portfolio
   - Check duration alignment with investor's interest rate view

- Assess sector concentration risks
- Evaluate tax implications for specific investor types

5 Python Implementation

5.1 Data Collection from SmartLab

Python code parses these indicators from SmartLab:

1. Лет до погашения
2. Доходность
3. Годовой купонный доход
4. Купонный доход последний
5. Рейтинг
6. Объем, млн руб
7. Купон, руб
8. Частота, раз в год
9. НКД, руб
10. Дюрация, лет
11. Цена
12. Дата купона
13. Размещение
14. Погашение
15. Оферта

5.2 Calculation Algorithms

Implementation includes these calculation functions:

1. Basic valuation: Standard discounted cash flow
2. Frequency adjustment: For bonds with non-annual payments
3. Reinvestment version: Including coupon reinvestment income
4. Tax-adjusted version: Considering investor taxes*; * - only 13% in the first versions
5. Yield calculations: Prediction of possible yield scenarios (positive, neutral, negative), depending on the CB Key Rate change trajectory*; * - will be added in later versions

5.3 Integration with Tinkoff API

The system connects to Tinkoff API for:

- Real-time prices: Current market prices for accurate comparison
- Trading data: Volume and liquidity information*; * - in later versions
- Historical data: For trend analysis if needed*; * - in later versions

For each bond basket, the system calculates:

- Average market price
- Average yield to maturity
- Standard deviation of prices and yields
- Price and yield ranges
- Duration-weighted average yield
- Yield curve parameters for the basket

5.4 Output Generation

The system produces these outputs:

1. Comparison tables: Sortable tables with price and yield deviations
2. Color coding: Visual indicators of valuation status and additional risks
3. Summary statistics: For each bond basket including yield curves
4. Anomaly lists: Separate lists for price and yield anomalies
5. Risk assessment: Credit risk and other risk factors
6. Investment recommendation: Categorized by time horizon (short/long-term)

6 Results and Analysis

TODO

7 Comparing Theoretical and Market Values

TODO

8 Conclusion

TODO

As financial markets continue their rapid automation (and development, speaking about our Russian stock market), tools like this will become highly important for investors seeking to maintain competitive advantages. This project demonstrates how algorithmic approaches can be applied to bond market analysis, creating opportunities for improved investment decisions.

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10 Appendices

10.1 Technology Stack

The project is implemented using the following technology stack:

1. **Data Collection (Parsing)**

- **Python + requests + BeautifulSoup4** — parsing static bond data from SmartLab (bond characteristics, coupons, ratings, maturity dates)
- **Tinkoff Investments SDK + gRPC** — real-time price data via Tinkoff API
- **pymoex + requests** — official data from Moscow Exchange (MOEX ISS API)
- **requests + XML/JSON** — macroeconomic data from Central Bank of Russia (key rate, inflation, currency rates)

2. **Data Processing and Storage**

- **pandas + numpy** — data cleaning, transformation, and analysis
- **PostgreSQL** — primary database for structured information storage
- **SQLAlchemy** — ORM for Python database interaction
- **APScheduler** — task scheduler for regular data updates

3. **Calculation Engine and Analysis**

- **numpy + scipy** — mathematical calculations (equation solving for YTM, numerical methods)
- **scikit-learn** — machine learning for anomaly detection (Isolation Forest), clustering
- **statsmodels** — regression analysis for yield curve construction
- **custom Python functions** — implementation of bond valuation formulas (discounting, duration, deviations)

4. **Backend API**

- **FastAPI** — web framework for REST API creation
- **Pydantic** — data validation and DTO (Data Transfer Objects)
- **Uvicorn** — ASGI server for application deployment
- **python-jose + passlib** — authentication and security (optional)

5. Frontend (Website)

- **React.js** — main framework for user interface
- **AG Grid** — interactive tables with sorting, filtering, and color coding
- **Chart.js** + **react-chartjs-2** — graphs (yield curves, data visualization)
- **Bootstrap** + **React-Bootstrap** — styling and responsive design
- **Axios** — HTTP client for backend requests
- **React Router** — page navigation

6. Infrastructure and Deployment

- **Docker** + **Docker Compose** — containerization of all services (DB, backend, frontend)
- **Nginx** — web server for frontend static files
- **Git** + **GitHub** — version control and code hosting

7. Documentation

- **Overleaf (LaTeX)** — academic documentation formatting
- **Swagger/OpenAPI** — automatic API documentation (built into FastAPI)
- **Markdown** — GitHub README

System Architecture Diagram:

```
SmartLab/Tinkoff/MOEX → Python parsers → PostgreSQL → Calculation Engine (Python) →
       FastAPI → React frontend → User
                          ↑                     ↑
                      Scheduler             scikit-learn
                      (APScheduler)          (ML models)
```

Project Repository:

All source code for this project is available on GitHub at the following repository:

`https://github.com/axiosdoom/Oblivion.git`

10.2 Example Output Tables

TODO